

Sample Exam Questions

Spatial Data Management

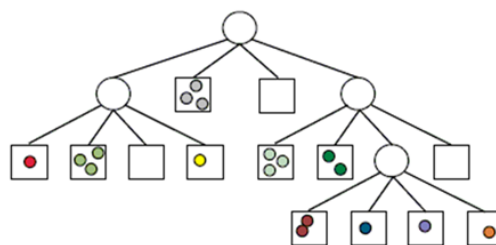
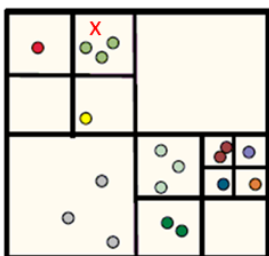
PART A

1. Why is storing the exact geometric extent of spatial objects often avoided?
 - A. It reduces accuracy
 - B. It increases storage and query processing cost
 - C. It violates relational normal forms
 - D. It prevents indexing
2. Representing roads as sequences of line segments is an example of:
 - A. Raster representation
 - B. Field modeling
 - C. Vector approximation
 - D. Grid decomposition
3. A GeometryCollection in spatial databases is used to:
 - A. Store raster tiles
 - B. Combine multiple geometry types in one object
 - C. Store multivalued numeric data
 - D. Enforce topology rules
4. Which of the following is a topological relationship?
 - A. Near
 - B. North of
 - C. Intersects
 - D. k-Nearest
5. Which query finds all features located inside a given polygon?
 - A. Range query
 - B. k-nearest neighbor query
 - C. Topological query
 - D. Spatial join
6. Which spatial query is computationally most expensive without indexing?
 - A. Attribute filtering
 - B. Range query
 - C. Point query
 - D. Distance-based join
7. The most used spatial approximation for filtering is:
 - A. Convex hull
 - B. Grid cell
 - C. Minimum Bounding Rectangle (MBR)
 - D. Centroid

8. Why is spatial indexing fundamentally harder than 1D indexing?
 - A. Spatial data cannot be stored on disk
 - B. There is no total order preserving spatial proximity
 - C. Spatial objects are always disjoint
 - D. Spatial objects are immutable
9. One limitation of space-filling curves is that:
 - A. They cannot be indexed
 - B. They do not preserve spatial locality perfectly
 - C. They only work for points
 - D. They require raster data
10. An R-tree indexes spatial data based on:
 - A. Exact geometry shapes
 - B. Minimum Bounding Rectangles (MBRs)
 - C. Raster cells
 - D. Spatial predicates
11. For which type of data distribution does the grid file perform best?
 - A. Highly skewed distributions
 - B. Dynamic and frequently updated datasets
 - C. Uniformly distributed static point data
 - D. Large objects with extent
12. Why can spatial range queries be transformed into multiple 1D range queries when using space-filling curves?
 - A. Because spatial objects are clipped
 - B. Because the curve defines disjoint partitions
 - C. Because the curve maps 2D regions to intervals on a 1D ordering
 - D. Because points are evenly distributed

PART B

1. Explain the difference between the topological relationships inside, adjacent and overlaps, using interior and boundary concepts.
2. Explain how Minimum Bounding Rectangles (MBRs) are used in spatial query evaluation. Illustrate your explanation with a concrete graphical example.
3. Consider the following graphical representation of a Quadtree, where the maximum capacity of each node is 3 points. Show the necessary transformations to insert into the Quadtree the point denoted by the red cross. Explain briefly your solution.



4. Explain dead space, object clipping, and overlapping partitions in spatial indexing with R-Trees and describe how each one influences query performance.